



Master's Degree Thesis Proposal Form

(To be prepare by the student and approved the supervisor)

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Thesis Title

a. As in the Language written by:

“Co-Encapsulation of Vorinostat and Vanillic Acid–
Derived Copper Nanoparticles into Liposomes for
Enhanced Cancer Therapy”

b. As translated to Arabic or English:

التغليف المشترك لفورينوستات وجزيئات نانوية من النحاس المشتقة من حمض الفانيليك في الليبوزومات لتعزيز توصيلهم لعلاج السرطان

- Graduate Studies committee chairman in the Department:.....Signature:
- Graduate Studies committee chairman in the faculty:.....Signature:





Part One: (Introduction)

Cancer continues to be one of the significant global health issues, accounting for nearly one in six deaths around the world (Sung et al., 2021). In 2019, there were about 18.1 million new cancer diagnoses and 9.5 million fatalities reported, with the incidence expected to rise by 40–50% by 2040 (WHO, 2020; Siegel et al., 2020). Cancer develops from abnormal and uncontrolled cell growth that can invade nearby tissues and spread to distant organs (Hanahan & Weinberg, 2011). Traditional chemotherapy remains the primary treatment for numerous cancers; however, it frequently lacks specificity and inflicts considerable toxicity on normal cells, resulting in severe side effects and diminished quality of life for patients (Chabner & Roberts, 2005; Jain, 2005). Consequently, there is an urgent necessity to create alternative therapeutic strategies that offer improved selectivity and reduced systemic toxicity (Allen & Cullis, 2013). Epigenetic therapy has surfaced as a promising approach in cancer treatment, as it focuses on reversible changes in gene expression without modifying DNA sequences (Dokmanovic et al., 2007). Among the epigenetic modulators, histone deacetylase inhibitors (HDACi) have demonstrated significant anticancer efficacy (Minucci & Pelicci, 2006). Vorinostat (suberoylanilide hydroxamic acid; SAHA) was the first HDAC inhibitor sanctioned by the U.S. FDA for the treatment of cutaneous T-cell lymphoma (Marks et al., 2001). Vorinostat functions by inhibiting zinc-dependent HDAC enzymes, which results in histone re-acetylation and the reactivation of tumor suppressor genes, thereby inducing cell-cycle arrest and apoptosis (Richon et al., 2000; Zhang et al., 2014). Nevertheless, its therapeutic effectiveness is constrained by poor aqueous solubility, rapid metabolism, low bioavailability, and the emergence of resistance (Kelly et al., 2003; Duvic et al., 2007). These pharmacokinetic limitations impede effective tumor





accumulation and lead to suboptimal clinical results (Karagiannis & El-Osta, 2006). To address these challenges, drug delivery systems (DDSs) have been thoroughly investigated to improve the solubility, stability, and targeted delivery of Vorinostat (Shi et al., 2017). Nanoscale carriers, including polymeric nanoparticles, micelles, and liposomes, facilitate extended drug circulation, enhanced bioavailability, and tumor-specific accumulation via the enhanced permeability and retention (EPR) effect (Maeda et al., 2013). Among these carriers, liposomes are particularly appealing due to their biocompatibility, biodegradability, and capacity to encapsulate both hydrophilic and hydrophobic drugs (Immordino et al., 2006; Bozzuto & Molinari, 2015). Liposomal formulations safeguard drugs from premature degradation, enhance pharmacokinetic profiles, and can be tailored for targeted delivery (Sercombe et al., 2015). Additionally, modifying the surface of liposomes with D- α -tocopheryl polyethylene glycol 1000 succinate (TPGS) improves drug solubility, stability, and cellular uptake, while also inhibiting P-glycoprotein (P-gp) efflux pumps to counteract multidrug resistance (Collnot et al., 2007; Zhang et al., 2014; Guo et al., 2020). TPGS-coated liposomes have shown enhanced permeability, extended half-life, and increased cytotoxic effects of loaded anticancer agents (Ruttala et al., 2017). Simultaneously, silver nanoparticles (AgNPs) have attracted considerable interest in cancer nanomedicine due to their distinctive physicochemical and biological characteristics, such as strong surface plasmon resonance, ease of functionalization, and inherent anticancer properties (Gurunathan et al., 2018; Singh et al., 2018). The biogenic synthesis of AgNPs utilizing plant extracts presents an eco-friendly alternative to chemical synthesis, resulting in improved biocompatibility and lower toxicity (Ahmed et al., 2016; Iravani, 2011). Plant secondary metabolites, including flavonoids, alkaloids, and phenolics, serve as natural reducing and stabilizing agents in the formation of nanoparticles (Mittal et al., 2013; Rai et al., 2013). *Tabebuia pallida*, belonging to the Bignoniaceae family, is particularly abundant in phenolic and





flavonoid compounds and has been effectively utilized for the green synthesis of AgNPs, which exhibit potential anticancer activity (Bhattacharjee et al., 2018; Adebayo-Tayo et al., 2019). Nevertheless, despite their promising applications, metallic nanoparticles frequently encounter challenges such as poor solubility in biological media and nonspecific biodistribution, which can lead to off-target effects (Wilhelm et al., 2016). The integration of biogenically synthesized AgNPs into liposomal carriers presents a synergistic approach that merges the therapeutic and diagnostic properties of AgNPs with the stability and controlled release capabilities of liposomes (Akbarzadeh et al., 2013). This hybrid nanoplatform has the potential to enhance tumor-specific accumulation through the EPR effect, promote endocytotic uptake, and allow for controlled intracellular release of therapeutic agents, thereby reducing systemic toxicity. In light of this rationale, the current proposal seeks to develop a TPGS-modified liposomal nano-system that encapsulates biogenic silver nanoparticles alongside Vorinostat. This multifunctional nanocarrier is intended to enhance Vorinostat's solubility, bioavailability, and intracellular delivery while leveraging the synergistic anticancer effects of silver nanoparticles. The anticipated outcome is a targeted, pH-responsive drug release mechanism within tumor microenvironments, providing an innovative therapeutic platform for more effective and less toxic cancer treatment.

A. Problem Statement:

Vorinostat (Suberoylanilide hydroxamic acid - SAHA), an HDAC inhibitor, is an effective anticancer agent, but its clinical use is often limited by poor bioavailability, lack of tumor specificity, and systemic toxicity. The research problem is to develop a multifunctional nanocarrier that can overcome these limitations by combining the drug with an active, stabilizing nanomaterial and an efficient delivery system.





B. Significance of the Study

This study presents a novel therapeutic strategy by integrating liposomal nanotechnology with a synergistic combination of bioactive components—Vorinostat, vanillic acid (VA), and copper nanoparticles (CuNPs). Vorinostat serves as the principal anticancer agent, while vanillic acid, a natural polyphenol, functions as a green reducing and capping agent, imparting antioxidant stability and additional anticancer synergy. The incorporation of copper nanoparticles is intended to potentiate cytotoxic efficacy through enhanced oxidative stress within tumor cells.

Encapsulation within liposomes not only protects these active components from premature degradation but also enables passive tumor targeting via the Enhanced Permeability and Retention (EPR) effect. This integrated system is expected to improve drug stability, bioavailability, and therapeutic selectivity, offering a promising platform for advanced cancer treatment.

C. Study Objectives

1. To successfully synthesize and characterize Vanillic Acid-Capped Copper Nanoparticles (Cu NPs/VA).
2. To formulate and optimize Vorinostat and Cu NPs- loaded Liposomes.
3. To determine the physicochemical properties of the final formulation (size, zeta potential, and encapsulation efficiency).
4.   Evaluate the in vitro cytotoxicity and apoptotic induction of the optimized  to see Vorinostat in relevant cancer cell lines.

D. Hypotheses and / Research Questions

Hypothesis: The encapsulation of CuNPs/VA and Vorinostat within liposomes will result in a stable nanocarrier with significantly enhanced cytotoxic and pro-apoptotic effects on cancer



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cells compared to free Vorinostat. Research Question: How does the synergistic action of Vanillic Acid-Capped Copper Nanoparticles and Vorinostat, when delivered via liposomes, affect the viability and mechanism of death in cancer cells?

E. Conceptual and Operational Definition of Variables (if any)

- Conceptual Variable: Therapeutic Efficacy.
- Operational Variables: Cell Viability (measured by MTT assay), Apoptosis Rate (measured by Annexin V/PI staining), Nanoparticle Size and Charge (measured by DLS/Zeta Potential).

Part Two: (Literature Review)

Liposomes were first discovered in the 1960s by British hematologist Dr. Alec D. Bangham and his colleagues at the Babraham Institute, University of Cambridge, with the initial report published in 1964 (Bangham and Horne, 1964). Liposomes are characterized as colloidal spherical structures that form through the self-assembly of amphiphilic lipid molecules in solution, such as phospholipids (Sebaaly et al., 2016). The membrane of liposomes can consist of one or more lipid bilayers (lamellas) arranged around an internal aqueous core, with the polar head groups oriented towards both the inner and outer aqueous phases (Nisini et al., 2018). This organized structure endows liposomes with the distinctive capability to load and deliver molecules with varying solubility. Hydrophilic molecules are contained within the internal aqueous core, hydrophobic molecules are embedded in the lipid bilayer, and amphiphilic molecules are positioned at the interface of the water and lipid bilayer (Laouini et al., 2012a). To this day, liposomes have been extensively studied in various pharmaceutical research contexts as drug delivery systems and remain a significant area of investigation (Bozzuto and Molinari, 2015). Liposomes are regarded as effective drug delivery systems due to their structural versatility, biocompatibility, biodegradability, and their non-toxic and non-



immunogenic properties (Mathiyazhakan et al., 2018). The amphiphilic nature of phospholipids in solution mimics natural cell membranes, facilitating excellent interactions between liposomes and mammalian cell membranes, which promotes efficient cellular uptake (He et al., 2019). Additional benefits of liposomes include their capacity to carry substantial drug payloads, their ability for self-assembly, and a broad spectrum of physicochemical and biophysical properties that can be tailored to regulate their biological characteristics (Sercombe et al., 2015). Liposomes are regarded as highly effective drug delivery systems due to their structural adaptability, biocompatibility, biodegradability, and their non-toxic and non-immunogenic properties (Mathiyazhakan et al., 2018). Furthermore, liposomes possess additional benefits such as the capacity to transport substantial drug payloads, the ability for self-assembly, and a diverse array of physicochemical and biophysical properties that can be tailored to regulate their biological characteristics (Sercombe et al., 2015). As a drug delivery system, liposomes have advanced therapies across various biomedical applications by stabilizing therapeutic agents, addressing barriers to cellular and tissue uptake, and enhancing the bio-distribution of compounds to target sites in vivo (Ding et al., 2006, Hua and Wu et al., 2013). Doxil (liposomal doxorubicin): The first nano-drug approved by the FDA in 1995, it is used for ovarian cancer. The drug encapsulated within the liposome is shielded from physiological occurrences, including enzymatic degradation, chemical and immunologic inactivation, and rapid plasma clearance, which contributes to the improvement and prolongation of its therapeutic action. Since the drug is contained within the liposome, its exposure to healthy tissues minimized, thereby reducing the adverse side effects associated with the free drug form (Bozzuto and Molinari et al., 2015). Liposomes are viewed as promising and versatile drug carriers. In comparison to conventional drug delivery systems, liposomes demonstrate superior characteristics such as targeted delivery, sustained or controlled release, protection





from degradation and clearance, enhanced therapeutic efficacy, and reduced toxic side effects. In light of these advantages, numerous liposomal drug formulations have been successfully approved and utilized in clinical settings over the past few decades. This review discusses the liposomal drug products that have received approval from the U.S. Food and Drug Administration (FDA) and the European Medicines Agency (EMA). It is based on the published approval documentation from the FDA and the European public assessment report (EPAR) from the EMA, which introduces critical chemistry information and advanced pharmaceutical technologies utilized in the marketed liposomal products. This includes details on lipid excipients, manufacturing techniques, nanosizing methods, drug loading strategies, and the critical quality attributes (CQAs) of these products. Furthermore, the current regulatory guidelines and future outlook regarding liposomal products are summarized. This information can be leveraged for the research and development of liposomal drug candidates across various stages, including laboratory bench work, pilot plant operations, and commercial manufacturing (Liu et al., 2022). Liposomes are self-assembled vesicles made from phospholipids that can form either a single bilayer (unilamellar) or multiple concentric bilayers (multilamellar) that enclose a central aqueous compartment (Smith & Lee et al., 2020). The size of liposomes varies from 30 nm to the micrometer range, with the phospholipid bilayer measuring approximately 4–5 nm in thickness (Jones et al., 2019). The field of liposomology was pioneered by British scientist Alec Bangham and his colleagues at Babraham, Cambridge, during the mid-1960s, who first published the structure of liposomes in 1964 (Bangham & Horne et al., 1964). Since that time, liposomes have been thoroughly researched as delivery systems for small-molecule drugs, proteins, nucleic acids, and imaging agents (Chen et al., 2018; Patel & Singh et al., 2020; Zhang et al., 2021). Various administration routes have been developed to improve therapeutic efficacy and patient compliance, including parenteral,



pulmonary, oral, transdermal, ophthalmic, and nasal routes (Huang et al., 2019; Kumar & Gupta et al., 2021). Additionally, liposomes are extensively utilized in the food industry (Li et al., 2020) and the cosmetic sector (Park & Kim et al., 2021). As drug delivery systems, liposomes present significant benefits, including safeguarding encapsulated materials from physiological degradation (Nguyen et al., 2017), extended circulation duration, regulated drug release (Wang & Zhao et al., 2018), and outstanding biocompatibility and safety. Furthermore, liposomes can target diseased tissues selectively through passive and/or active targeting, which reduces systemic side effects and enhances therapeutic results (Singh et al., 2019; Zhao et al., 2020). In contrast to normal tissues that possess tight intercellular junctions (2–6 nm) between endothelial cells (Anderson & Lee et al., 2018), abnormal tissues, such as solid tumors, display highly permeable capillaries (100 nm–2 μ m, depending on the type of tumor) (Miller et al., 2020). Liposomes are capable of infiltrating these leaky blood vessels and accumulating at tumor locations, a phenomenon referred to as the enhanced permeability and retention effect (Maeda et al., 2013). Active targeting utilizes specific interactions between ligands on the liposome surface and receptors that are overexpressed on tumor cells, including VEGF, EGF, folate, integrins, CD44, CD13, and PSMA (Xu et al., 2019). A variety of ligands—such as antibodies (Chen et al., 2017), nucleic acids like aptamers (Liu & Zhang, 2021), proteins such as transferrin (Wang et al., 2020), peptides like iRGD and iNGR (Johnson & Patel et al., 2019), small molecules like folic acid (Lee et al., 2020), and carbohydrates such as dextran and mannose (Huang et al., 2018)—have been suggested for the surface modification of liposomes. Despite these benefits, only approximately 14 liposomal drug products are presently available on the market, indicating that the complete potential of liposomes has yet to be fully realized (FDA, 2022). Consequently, this review consolidates information on commercial liposomal products that have received approval from the FDA and EMA,



concentrating on their composition, manufacturing technologies, critical quality attributes (CQAs), and regulatory viewpoints, to support future development of liposomal drugs (Liu et al., 2022). Liposome–inorganic nanoparticle hybrids have emerged as sophisticated multifunctional drug delivery systems that combine the biocompatibility and controlled-release characteristics of liposomes with the diagnostic and therapeutic capabilities of inorganic nanomaterials (Liu et al., 2022). Liposomes, which consist of self-assembled phospholipid bilayers, are capable of encapsulating both hydrophilic and hydrophobic drugs, thereby enhancing their stability, bioavailability, and targeted delivery (Bangham & Horne, 1964; Allen & Cullis, 2013). Nevertheless, conventional liposomal formulations often encounter challenges such as inadequate imaging capabilities and limited control over drug release. To address these issues, researchers have integrated inorganic nanoparticles, including gold, iron oxide, mesoporous silica, and quantum dots, into or onto liposomes, resulting in hybrid systems with synergistic properties (Chen et al., 2018; Zhang et al., 2021). Gold nanoparticles (AuNPs) and gold nanorods are commonly utilized due to their exceptional photothermal conversion efficiency, which facilitates light-triggered drug release and the combination of chemo-photothermal therapy (Huang et al., 2020). Iron oxide nanoparticles (Fe₃O₄) provide magnetic responsiveness, enabling magnetic targeting and magnetic resonance imaging (MRI) for real-time monitoring of drug biodistribution (Patel & Singh, 2021). Mesoporous silica nanoparticles, when either encapsulated or coated with



liposomes, present a high surface area and adjustable porosity for substantial drug loading and controlled release (Li et al., 2020). Likewise, upconversion and quantum dot-based hybrids offer fluorescence and photoactivation features that are beneficial for imaging and photodynamic therapy (Wang et al., 2019). These hybrid systems facilitate external control over drug release through thermal, magnetic, or light stimuli, thereby enhancing spatial and temporal precision in treatment (Maeda et al., 2013; Zhao et al., 2020). Despite their potential, liposome–inorganic nanoparticle hybrids continue to encounter obstacles concerning large-scale production, reproducibility, long-term biocompatibility, and regulatory approval (Singh et al., 2019). The possible toxicity of inorganic cores, particularly those derived from heavy metals, requires thorough assessment of degradation and clearance profiles (Anderson & Lee, 2018). Nevertheless, recent research indicates that these hybrid systems can markedly improve tumor accumulation, therapeutic effectiveness, and imaging contrast when compared to traditional liposomes (Zhang et al., 2021; Liu et al., 2022). In summary, liposome-encapsulated inorganic nanoparticles signify a promising next generation of nanocarriers for targeted drug delivery and theranostics, although further enhancement of safety, stability, and scalability is crucial for clinical application (Chen et al., 2018; Patel & Singh, 2021).

Part Three: (Study Methodology)

A. Design

- 1- Synthesis of VA-CuNPs.
- 2- Formulation and optimization of lecithin–cholesterol liposomes co-loaded with Vorinostat and VA-CuNPs.
- 3- Physicochemical characterization and in vitro biological evaluation of the developed formulation.



4- Setting

An aqueous solution of vanillic acid (5 mM) will be prepared and adjusted to PH. Separately, an aqueous solution of copper (II) sulfate pentahydrate (1–5 mM) will be prepared. The copper solution will be added dropwise to the vanillic acid solution under constant stirring.

The mixture will be stirred for 1–2 hours until a visible color change from blue to brown is observed, indicating the formation of VA-CuNPs. The resulting nanoparticles will be collected by centrifugation at 12,000 rpm for 15 minutes and washed three times with deionized water to remove unreacted components. The purified VA-CuNPs will then be redispersed in deionized water and stored at 4 °C in the dark.

For liposomes preparation, lecithin and cholesterol will be accurately weighed in a molar ratio of 7:3 and dissolved in organic solvent. Vorinostat will be added to the lipid solution at a drug-to-lipid ratio ranging from 1:10 to 1:5 (w/w). then we will be synthesized the loaded liposomes using thin film hydration method or ethanol injection based on preparation optimization.

5- Population of the Study and Sample (if any)

NA

6- Study Instruments

- Controlled heating and stirring, Cu is not magnetic but it can form some weak repulsive forces when exposed to a magnetic field.
Some nice to study is and think about is to do the heating and vortexing in Thermoshaker vs stirring plate with magnets and notice the effect.
- Centrifuge – for separation and purification of nanoparticles.
- The ratio that have been bound to vanillic acid from the copper, is it based on titration or it will be based on analytical UV measurements in the HPLC?
Fourier Transform Infrared Spectroscopy (FTIR) – to identify functional groups and confirm capping of CuNPs by vanillic acid.



- X-Ray Diffraction (XRD) – to determine crystalline structure and particle size of CuNPs.
- Dynamic Light Scattering (DLS) Analyzer – to measure particle size distribution and polydispersity index (PDI).
- Zeta Potential Analyzer – to determine surface charge and stability of nanoparticles.
- Imaging via staining or using Cryo imaging. Cryo will be more realistic since water will form a thin layer of ice surrounding your liposomes and allow contrast to happen as well we better resolution than staining.
- Rotary Evaporator – for thin-film hydration method to form liposomes.
- Probe Sonicator – for size reduction and uniform dispersion of liposomes.
- pH Meter – for adjusting hydration medium pH.
- Vacuum Desiccator – for solvent removal and drying of lipid films.
- High-Speed Centrifuge – for separation of free and encapsulated drug/nanoparticles.
- UV–Visible Spectrophotometer or HPLC – for encapsulation efficiency and drug release analysis.
- Incubator (CO₂ and Humidity Controlled) – for maintaining cell culture conditions.
- Laminar Flow Hood – for aseptic cell handling.
- Inverted Microscope – for observing cell morphology.
- Microplate Reader (ELISA Reader) – for MTT cytotoxicity assay to study the effect of copper, vanillic acid alone on the cells and drug release kinetics.
- Centrifuge and Vortex Mixer – for sample preparation and mixing.
- Analytical balance
- Glassware (flasks, beakers, pipettes, etc.)
- Refrigerator and deep freezer (for storage of reagents and samples)
- Water bath and heating mantle
- Filtration unit (Whatman filters or syringe filters 0.22 µm)

7- Ethical Considerations

NA

8- Data Collection Process

NA



9- The Plan for Data Management Analysis

All experimental observations and measurements will be systematically recorded in a laboratory notebook and supported by corresponding digital records. Each sample and experiment will be assigned a unique identification code to ensure traceability and proper documentation. All collected data will then be analyzed using Microsoft Excel and/or GraphPad Prism to obtain statistical parameters, graphical representations, and comparative analyses.

10- Other Remarks Specific for this Thesis

Part Four: References (APA Style)

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